When 100% Acceptance Means One Allowed Price

Interface-dependent discretion in guarded Q-learning markets

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Abstract

A price guard can protect consumers while replacing most decisions made by a pricing agent. Does a low correction rate show that the agent retains economic discretion? We study this question with independent, own-profit tabular Q-learners in one differentiated-product duopoly. A prospectively specified initial study finds that a profit-tolerance guard raises consumer surplus from 45.29 to 72.11 over 20,000 adaptive rounds, but retains only 6.75% of proposals. A prospectively specified follow-up compares proposal credit, naive executed-action credit, and feasible-action-aware Q-learning, with matched price-cap and unguarded controls. Across twelve original checkpoints, guard-aware learning raises retention to 100% while leaving welfare nearly unchanged and permitting approximately one of fifteen prices. Equal-budget training on twelve new seeds yields identical protected execution and zero price entropy across guard variants. Removing protection produces no demonstrated welfare advantage for the aware variant. A 100,000-round continuation also shows that the cap's early welfare advantage largely vanishes. These results distinguish a proposal-interface metric from feasible choice and protected performance from unconstrained behavior. The contribution is a reproducible economic comparison using established learning and shielding methods, not a new anti-collusion algorithm or evidence about language-model intent.

1 Introduction

Pricing agents can learn persistent high prices even when each maximizes only its own profit [1,2]. An external guard offers a tempting remedy: let the agent propose any price, then execute a price close to its immediate best response. Consumer benefit can improve, but this alone does not show that the agent retains useful choice or learns competitive behavior without enforcement.

Our question is narrow: **does apparent pricing discretion survive changing the proposal interface, and what economic behavior survives removal of the guard?** We compare matched saved policies, actual alternative learners, simple controllers, and a longer continuation. The result is a caution about evaluation: a perfect acceptance score can coexist with only one feasible price. Learned price proposals make this an empirical study; the elementary projection identity alone is not its scientific contribution.

2 Related Work

Calvano et al. [1] establish the use of tabular Q-learning to study algorithmic pricing. Eschenbaum, Mellgren and Zahn [2] examine deployment with unfamiliar counterparts and distinguish frozen from adaptive interaction. Our learning rules and market are adaptations of this literature; we do not claim a faithful execution of either complete replication package. Independently seeded runs vary initialization and exploration, not the underlying economic calibration.

Shielding [3] already compares proposal credit, executed-action credit and preemptive action restriction; its Section 5.2 explicitly anticipates rejected proposals remaining optimal and the need for persistent enforcement. Projection learning and aliasing are also established [4,5]. Feasible-action selection and a feasible Bellman maximum are standard repairs. The sprint starter [6] also includes work on pricing agents and replacement. This report adds a matched economic comparison of proposal bookkeeping, feasible price support, welfare, seller profit and release. We found no basis for a first-ever aliasing, shielding, handover or collusion claim.

Platform-level welfare interventions against Q-learning sellers are also established [8]. Our lagged one-step modeled profit-loss constraint differs from calibrated-regret auditing [9]: it is neither a legal collusion test nor a calibrated-regret certificate. No superiority over these broader approaches is claimed.

3 Methods

3.1 Economy, private learners and timing

Two firms sell differentiated goods to 100 potential customers, with quality 2, unit cost 1, logit scale 0.25 and outside utility zero. With $z_i = \exp((2 - p_i)/0.25)$, demand is $q_i = 100\, z_i / (1 + z_A + z_B)$, profit is $(p_i - 1)q_i$, and consumer surplus is $CS = 25 \log(1 + z_A + z_B)$. CS is a model quantity. The 15-price grid spans 1.4277 to 1.9702; the lower symmetric grid-Nash reference is 1.46647. The grid has two symmetric pure Nash pairs.

Each firm observes the previous executed price pair, keeps its own Q table and random stream, and receives only its own realized profit. Both current actions are sealed before settlement. Training uses three million simultaneous rounds, alpha = 0.15, gamma = 0.95 and exploration $\exp(-0.000004\, t)$. Initial action values use discounted profits against uniform random rival prices; ties choose the lowest index. No cooperation reward, monopoly target or programmed retaliation is supplied. Q(s,a) is updated toward r + gamma max Q(s_next,b).

3.2 Controllers, learning interfaces and study units

For previous rival index j, the guard permits prices whose own one-round profit is at most 0.1 below the grid best response. The nearest permitted price executes. A simple cap instead clips to the grid price nearest continuous Nash, permitting two grid prices. We also retain uncontrolled, exact best-response and wider-tolerance controls.

Proposal credit selects among all 15 actions and credits the proposal. **Naive executed credit** credits the executed index but still selects and bootstraps over all actions, including infeasible cells; this is a potentially misspecified diagnostic. **Guard-aware learning** restricts selection, exploration and the next-state maximum to the appropriate feasible set, crediting the executed action.

The original study uses 12 seeded runs (1001-1012); adaptive evaluation lasts 20,000 rounds at exploration 0.01, while frozen evaluation lasts 2,000 without updates or exploration. The prospectively specified follow-up reuses every original checkpoint, trains five paired variants on 12 new seeds (4001-4012), and adds 100,000-round continuations from original checkpoints. All-seed paired bootstrap intervals resample whole runs 10,000 times. Branches and rounds do not increase n. Appendix A records provenance and scope.

4 Results

4.1 The original guard helps customers but fails its stated gate

All 12 original runs meet the prespecified above-reference price/profit and lower-CS screen. This establishes persistent harmful pricing under this protocol, not deliberate collusion. A memoryless sensitivity also learns high prices, so high prices alone cannot diagnose a strategic punishment mechanism. Every original run remains in the treatment comparison.

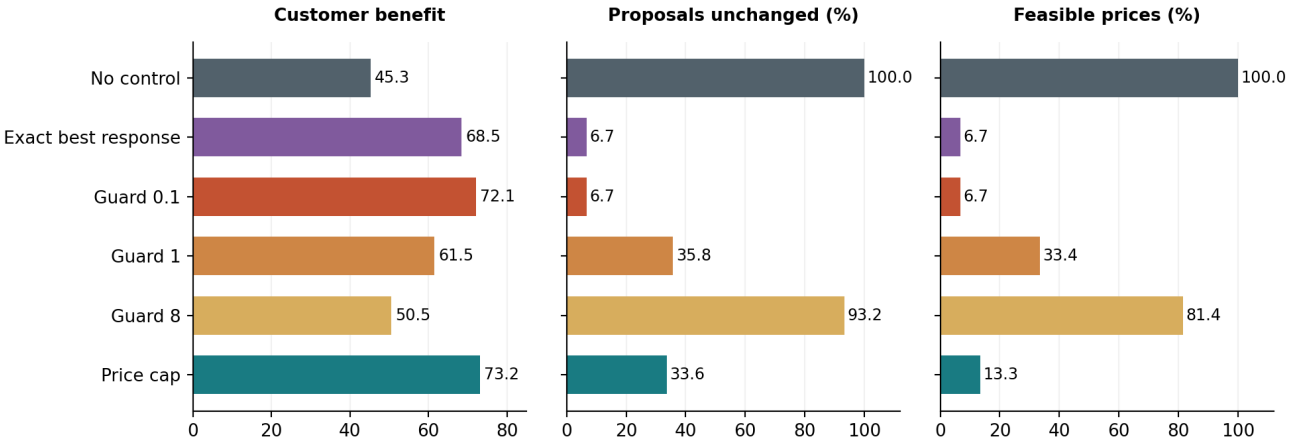


Figure 1. Original 20,000-round adaptive means across 12 paired seeded runs. Consumer benefit, proposal retention and feasible executed-price fraction are distinct outcomes. The guard tolerance is a one-round profit-loss allowance; the cap is a separate controller.

Table 1. Original adaptive means; feasible counts average both firms and time.

Controller	CS	Profit / seller	Retained	Feasible / 15
No control	45.292	31.322	100.00%	15.000
Exact best response	68.477	23.628	6.71%	1.000
Guard 0.1	72.114	22.022	6.75%	1.001
Guard 1	61.499	26.418	35.76%	5.008
Guard 8	50.531	29.801	93.25%	12.217
Price cap	73.187	21.525	33.63%	2.000

The prespecified guard-minus-uncontrolled paired CS gain is 26.8226, with 95% bootstrap interval [25.3820, 28.3111]. It exceeds the required 1.940098 CS units (5% of the continuous Nash-to-joint-monopoly CS gap); the 20% retained-proposal threshold fails at 6.7452%. The original combined-gate result remains negative. However, the follow-up shows why this retention threshold cannot establish interface-independent autonomy.

Over this short window the cap gives higher CS and more retained proposals than guard 0.1, but lowers average seller profit from about 22.022 to 21.525. It therefore does not dominate every objective. Section 4.3 also shows that its welfare lead is largely transient.

4.2 The learning interface changes retention without restoring choice

First, projection is idempotent at every one of the 225 prior-rival/action cells: M_j(M_j(a)) = M_j(a). Replacing recorded proposals by recorded executions preserves that economic path and gives 100% retention. This is an ex-post path reexpression. It does **not** preserve proposal-index Q updates or prove that an identically trained policy would result after relabeling. We therefore test actual guard-aware learners separately.

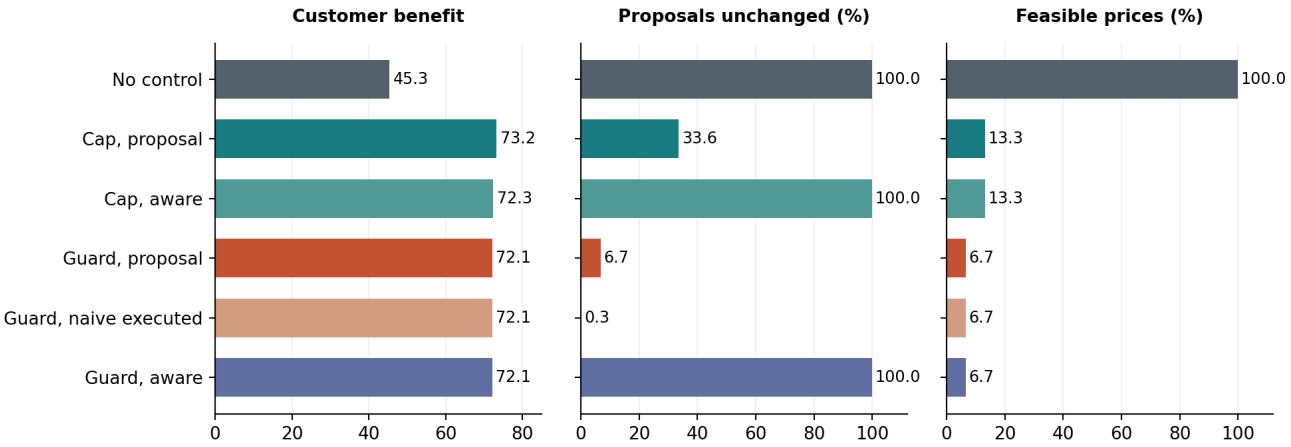


Figure 2. Actual alternative learners deployed from the same 12 original checkpoints, 20,000 adaptive rounds. Guard-aware selection gives 100% retention while roughly one of 15 prices is feasible. Changing a correction meter is not equivalent to increasing feasible choice.

Table 2. Matched adaptive deployment. Span is the maximum minus minimum permitted price.

Actual interface	CS	Retained	Feasible / 15	Price span
Cap, proposal	73.1873	33.630%	2.0000	0.038748
Cap, aware	72.3427	100.000%	2.0000	0.038748
Guard, proposal	72.1143	6.745%	1.0011	0.000044
Guard, naive executed	72.1068	0.293%	1.0033	0.000126
Guard, aware	72.1177	100.000%	1.0002	0.000006

Guard-aware minus proposal-credit CS is 0.003431 [0.000012, 0.008063]. This small transition difference accompanies a 93.25 percentage-point retention increase, not a comparable economic gain. Mean feasible span is only about 0.000006, and executed-price entropy about 0.0028 bits. Entropy is marginal outcome diversity, not strategic autonomy.

At the shared terminal rival price 1.46647, the second-best grid price loses 0.110063 profit units, exceeding tolerance 0.1. The singleton set therefore follows directly from this grid/tolerance combination; it is not a general property of price guards.

The aware intervention changes information about feasibility, policy support and bootstrap support together. It is a practical interface comparison, not identification of credit assignment as the sole cause. The naive executed-credit diagnostic retains stale optimistic infeasible Q entries; its weak retention is not evidence that constrained Q-learning itself fails.

4.3 Longer adaptation and new training change the interpretation

Long continuations start again from all original checkpoints, not from selected successful branches. In the final 20,000 of 100,000 rounds, all three guard variants execute the same single price: CS 72.1190, seller profit 22.0204, zero feasible span and zero executed-price entropy. Proposal and naive executed credit retain 0.0725% of proposals; aware learning retains 100%. Thus the contrast persists beyond the original short adaptation window.

Table 3. Final 20,000 rounds of the 100,000-round continuation; n = 12 paired seeds.

Late continuation	CS	Retained	Feasible / 15	Entropy bits
No control	46.7921	100.000%	15.0	1.6745
Price cap	72.1217	8.411%	2.0	0.0081
Guard, proposal	72.1190	0.072%	1.0	0.0000
Guard, naive executed	72.1190	0.072%	1.0	0.0000
Guard, aware	72.1190	100.000%	1.0	0.0000

The cap-minus-aware CS lead falls from about 1.07 over the initial 20,000 rounds to 0.00261 in this late window. The remaining difference is economically small and compatible with the cap occasionally allowing a lower price during exploration. We therefore withdraw any claim of a substantial persistent cap welfare advantage. These finite windows still do not establish mathematical convergence.

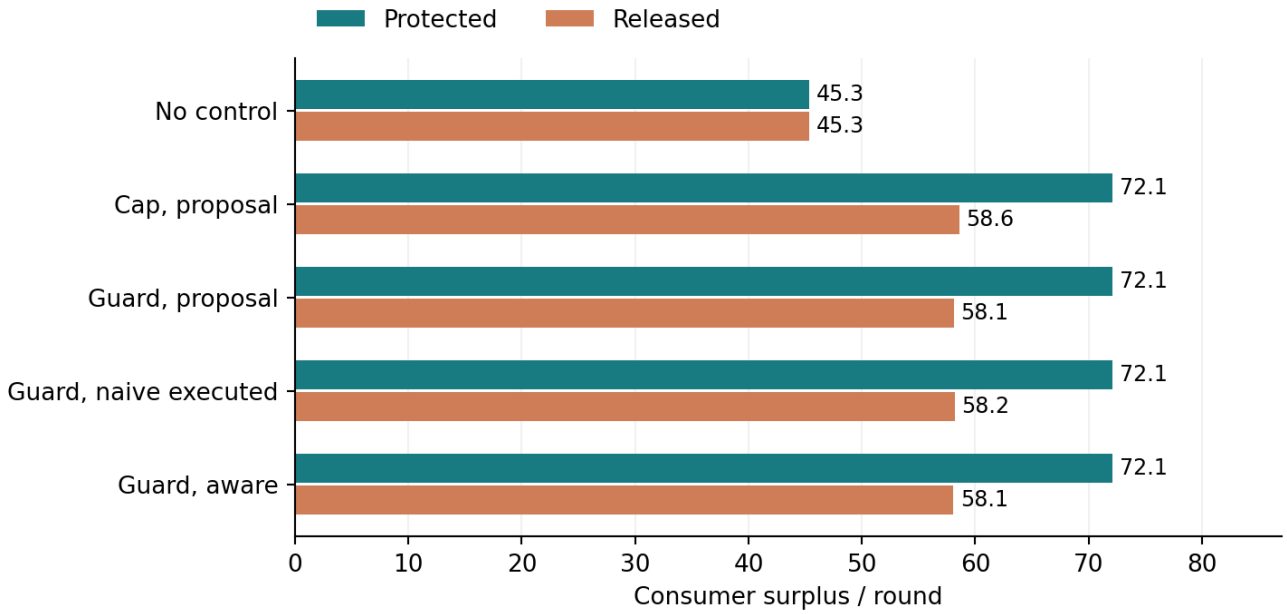


Figure 3. Twelve new paired seeds, five interfaces, three million training rounds per variant. Each protected/released pair begins from the identical terminal checkpoint and runs 20,000 adaptive rounds. Removal of both guard and mask reveals unconstrained behavior.

All three newly trained guard variants have protected CS 72.1190 and zero execution entropy; only the aware variant retains all proposals. Once released, aware CS is 58.0824 versus 58.0967 for proposal credit: paired difference -0.0144 [-1.0772, 0.7908]. This does not demonstrate a release benefit or equivalence. Unvisited initialized Q cells, changed action support and short adaptation remain alternative explanations.

5 Discussion and Limitations

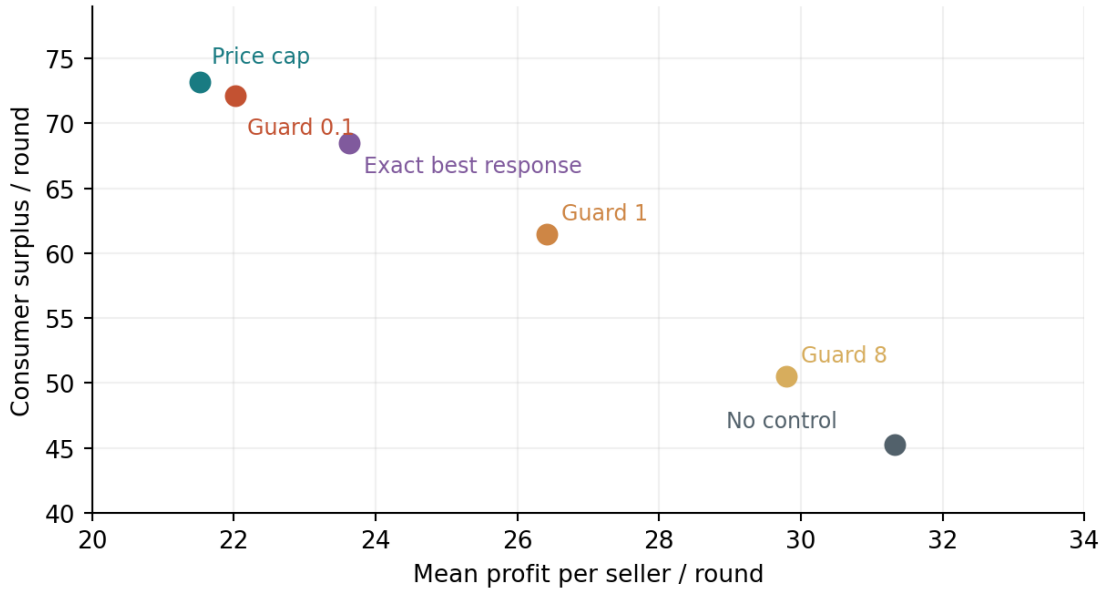


Figure 4. Original short-window comparison: consumer surplus and mean seller profit are competing objectives. Points are 12-run adaptive means, not a universal welfare frontier. The short-window cap advantage in customer benefit largely disappears in the longer test.

The useful finding is an evaluation distinction: full acceptance can coexist with one feasible price, while frequent corrections need not persist under another interface. Welfare or retention alone would miss this. Feasible support, price span, profits and release tests complement each other; none is a general autonomy score.

Limitations

This is one symmetric logit calibration with a finite grid, fixed hyperparameters and 12 seeded units per cohort. Multiple grid equilibria exist; the selected tolerance creates singleton support near the observed endpoint. Wider tolerances have different tradeoffs. Bootstrap intervals quantify seed variation, not real-market prevalence. Training is budget-bounded; the follow-up is motivated by original outcomes. The methods do not test language semantics, intentional coordination, strategic credibility, legal collusion or human comprehension. The separate earlier continuous-market derivation does not certify this grid learner. Numerical replay establishes consistency, not external validity.

Future Work

Next tests should vary grid and tolerance across economies, compare competent constrained learners, and test sustained release under new demand and counterparts. Semantic handover requires separate authorized language-agent experiments; more rounds here cannot answer it.

6 Conclusion

In these seeded pricing runs, changing the action interface can turn very low retention into 100% acceptance while leaving one allowed price and essentially identical protected economics. Guard-aware training does not show an unconstrained welfare benefit after release, and an apparent short-window cap advantage is largely transient. These bounded negative and diagnostic results support more careful controller comparisons; they do not establish a new broadly effective anti-collusion defense.

Code and Data

The accompanying local evidence archive includes frozen sources, seeds, checkpoints, compact action traces, all outcomes, analysis and independent auditors. Start with START-HERE.txt. The playable graphical demonstration is explanatory; its continuous human-price interface is distinct from the finite-grid experiment. No public code URL or external submission is claimed. Appendix A gives reproduction commands.

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Author Contributions

Mohammadpayam Teabi directed the project, provided its original game context, selected the research direction and authorized the local work. AI development agents assisted problem formulation, implementation, experimental execution, analysis, literature checks, review and report drafting. The report does not claim independent human replication or human-authored analysis where none occurred. See the LLM Usage Statement.

Template and attribution

This PDF follows the accessible official template's single-column research-paper structure, title rules, author block, abstract, numbered sections, references and LLM statement [7]. The required ethical appendix follows the newer event-page requirements. It is rendered locally rather than exported from the shared Google document. Established economic and learning ingredients are credited above; code provenance is detailed in the archive.

Appendix A. Limitations and Dual-Use / Ethical Considerations

A.1 What the result establishes and does not establish

The supported claim is a matched, finite-protocol comparison in a toy market. The original primary welfare/retention gate failed. The follow-up exposes its interface dependence rather than retroactively changing the outcome. The 12 reused checkpoint markets and their long branches are not new replications. Twelve new seeds are paired across five training variants; these 60 training markets contain 12 independent seed units, not 60 independent comparative replications. Late-window summaries reuse their parent 100,000-round paths. No population collusion rate, equilibrium theorem, universal shielding failure, intent detector or first-place prediction follows.

A.2 Provenance, controls and verification

The original protocol and prospective amendment were frozen before its substantive run. Its primary reference and gate were not retuned. New credit/aware experiments, source and checkpoint hashes, analysis and longer-continuation amendment were frozen separately before their respective new outcomes. Extension training comprises 180 million local rounds; its 384 short and 60 long branches total 10.224 million evaluated transitions. Every evaluated transition can be independently replayed from saved starting Q tables and PRNG state. Independent audit covers economics, projections, masks, random choices, evaluation updates, final checkpoints, bins and aggregates; it does not replay all training updates. Main-study audits and bounded ancillary checks have separate scopes.

Run the offline app with `node research/demo-server.js`. Run extension checks with `node --test --test-isolation=none tests/credit-learning.test.js`; the independent saved-data audit is `node research/audit-credit-study.js`. Regenerate tables with `node research/credit-analysis.js` and figures with `python research/build-sprint-figures.py`. Frozen original study records remain available separately. Exact rerun commands and dependencies are in START-HERE.txt.

A.3 Prior work, dual use and practical limits

The repository includes pre-sprint game work. The submitted tabular guard study, new credit/mask follow-up and report were developed during the sprint; the provenance ledger labels earlier material and established algorithms. The old semantic-handover and continuous-guard demonstrations are background, not evidence of language understanding. No new API game players or paid compute were used. Earlier exploratory account-backed pilots consumed 342 requests; those pilots are excluded from this report's evidence.

Multi-agent pricing research could help build or tune anticompetitive pricing systems, and misleading acceptance metrics could obscure restrictive behavior or weak oversight. We publish the limits, economic costs and ordinary baselines together with outcomes. This artifact contains a toy environment, not deployment instructions, customer data or real-market certification. No independent human playtest was conducted.

LLM Usage Statement

Codex research and coding agents assisted conceptualization, coding, local execution, analysis, literature review, adversarial feedback and drafting. This final text was substantially AI-drafted under the author's direction; it is not represented as primarily human-written. Reviews labeled independent are separate AI-agent implementation reviews, not human peer review. Claims were checked against saved results and primary sources, and implementation outputs were independently replayed as described. The tabular experimental sellers themselves made no LLM/API calls. The author remains responsible for the report and submission.